01 PROBLEM UNDERSTANDING AND OBJECTIVE

Clinical Document Intelligence Hub · 2-day prototype

Clinical and administrative staff read the same documents twice.

Intake forms, discharge summaries, lab reports and physician notes arrive as text, PDF, scans and faxes. Someone reads each one for the same handful of facts, then re-keys them elsewhere. Slow, inconsistent between readers, and the errors it produces are silent.

The hard part is not reading the document. It is whether a clinician can ***check*** what was read — quickly, per field, without re-reading the chart.

OBJECTIVE

Ingest one unstructured clinical document and return a decision-ready patient summary card:

- structured fields, each with a confidence and its source quote
- a risk flag a clinician can defend
- concrete recommended next steps

THREE CONSTRAINTS THAT SHAPED EVERY DECISION

Absent data must stay absent

Silent imputation is the failure mode that actually hurts someone. Missing fields become the literal string “not documented”; if three or more NEWS2 parameters are missing the band is INDETERMINATE, not a score built on partial vitals.

The risk score must not be generated

A prose rationale from a model cannot be tested, audited, or held still while a prompt is edited. NEWS2 is ordinary Python, covered by 18 tests.

Every value must be traceable

Each field carries a verbatim source quote, expandable in the UI. Verifying a dose should take a click, not a re-read.

Synthetic data only — no real PHI. Decision support, not a medical device.

02 SOLUTION ARCHITECTURE AND DESIGN FLOW

One model call, one schema, one deterministic scorer

INPUT
Text · PDF
Image · Scan
single document

INGEST
bytes + MIME type
src/ingest.py
no OCR stage

EXTRACT
Gemini 3.6 Flash
one call
schema-enforced

VALIDATE
ClinicalExtraction
Pydantic
typed or rejected

the typed record forks

DETERMINISTIC `src/triage.py`

NEWS2 risk score — plain Python

Royal College of Physicians scoring, computed from the extracted vitals. Every point is attributable to a reading. 18 tests pin the bands. The model never sees this step.

GENERATIVE `summary · reasoning · next steps`

Narrative, grounded in quotes

Prose a clinician reads in under 20 seconds, plus recommended actions ordered most urgent first and an explicit list of what is missing from the document.

The model extracts; Python decides. The risk band cannot drift when a prompt is edited, a clinician can be shown which reading contributed which point, and the whole scoring layer is testable without an API key.

Streamlit renders the summary card, extracted-detail tables, NEWS2 breakdown, source text and a JSON export.

03 IMPLEMENTATION HIGHLIGHTS

Four decisions, and the numbers they produced

ONE CALL — THE API ENFORCES THE SCHEMA

```
interaction = client.interactions.create(
    model=model,
    system_instruction=SYSTEM_PROMPT,
    input=_input_blocks(document),
    response_format={
        "type": "text",
        "mime_type": "application/json",
        "schema": response_schema(),
    },
    store=False,   # no clinical text retained
)
```

No JSON parsing, no repair loop, no prose-to-struct step that can drift. The response is constrained on the way out and re-validated by Pydantic on the way in — a malformed record raises rather than reaching the UI.

THE PROVIDER GOTCHA WORTH KNOWING

```
# Pydantic hoists nested models into $defs and
# points at them with $ref. Gemini documents no
# support for either keyword.

def response_schema() -> dict:
    schema = ClinicalExtraction.model_json_schema()
    defs = schema.pop("$defs", {})
    return _inline_refs(schema, defs)

# Inlined before the request goes out, rather
# than relying on undocumented behaviour.
```

Found by reading the SDK's own request types rather than trusting the docs page — which also revealed the API accepts TIFF, the format scanned clinical faxes actually arrive in.

MEASURED ON THE FIVE BUNDLED SAMPLES

85 / 85	0	31	~29 s	$0.00
source quotes traced back to the document	quotes containing fabricated content	tests — scoring and failure handling	per document, end to end	free tier — reproducible by anyone

Model: gemini-3.6-flash · extract.py is the only module aware of the provider.

04 CHALLENGES AND LEARNINGS

The confidence score was decoration — measuring it is what proved that

The prototype reported **high** confidence on almost every field. That looks like success until you ask whether the signal ***moves***. So I rendered the same ED triage note at three degradation levels and compared confidence against ground truth.

BEFORE confidence is flat while accuracy collapses

Degradation	Vitals right	Confidence
As shipped	6 / 6	21 high, 1 med
Degraded	5 / 6	**23 high, 0 med**
Severe	**3 / 6**	**20 high, 0 med**

AFTER the signal tracks legibility

Degradation	Vitals right	Confidence
As shipped	6 / 6	8 high, 16 med
Degraded	4 / 6	6 high, 18 med
Severe	1 / 6	**0 high, 17 med**

At severe degradation the model read temperature as 37.9 °C instead of 38.9, and heart rate as 118 instead of 124 — and labelled both **high** confidence.

The fix was conceptual, not mechanical: confidence had been written as a statement of **belief**. It is now a statement about the **source** — any digit that could be a different digit under this much blur is **medium**, however sure the model feels. Clean digital text is unaffected: 105 high / 0 medium.

ALSO LEARNED

The cheap model was a trap

flash-lite: 4x faster, same NEWS2, but it fabricated 3 of 65 source quotes — one carried the reference range from the row above. Rejected.

A green tick is a claim

Uniform high confidence tells a clinician there is nothing to check. That is worse than showing no confidence at all.

Read the SDK, not the docs page

$ref support and the real accepted MIME types were both only discoverable in the SDK's request types.

Degradation levels generated reproducibly (fixed seed) by scripts/make_scanned_sample.py. · The after-fix ladder reproduced exactly (8 → 6 → 0 high) on a second independent run.

DEMO SUMMARY AND NEXT STEPS

Five synthetic documents, two modalities, one pipeline

END-TO-END RESULTS AGAINST THE LIVE API

Document	Input	Classified as	NEWS2	Risk
ED triage note — suspected urosepsis	text	triage_note	**17**	HIGH
same note, printed → photocopied → faxed	**image**	triage_note	**17**	HIGH
Discharge summary — resolved sepsis	text	discharge_summary	0	ROUTINE
New-patient intake form — headaches	text	intake_form	0	ROUTINE
Lab report — potassium 6.8, no vitals recorded	text	lab_report	—	INDETERMINATE

A degraded fax produced the identical decision — **NEWS2 17, HIGH, all 7 parameters, all 7 labs, the same 5 flags.** No OCR stage.

The lab report has no vitals, so it is **not scored**. A NEWS2 of 0 would read as “patient is fine” when nobody took observations — the critical potassium still surfaces.

GIVEN MORE TIME

— **Close the audit trail.** One run produced a vitals quote that skipped a line while reading as continuous. Mark elisions; give each vital its own quote.

— **Compare across documents.** The brief's one optional enhancement still unbuilt: a discharge summary and a later lab report for the same patient should diff.

— **Pin the non-determinism.** Values are stable across runs; formatting is not (BD vs twice daily). Normalise in the schema.

TO TAKE THIS BEYOND A PROTOTYPE

— **Gate on patient category.** NEWS2 is validated for acutely ill adults — not paediatric or obstetric patients, and the tool does not detect those cases.

— **Human-in-the-loop by default.** Reviewed-and-accepted should be a recorded state with reviewer and timestamp, not an assumption.

— **Structured interoperability.** Emit FHIR resources rather than bespoke JSON, so output can re-enter the record it came from.